The "Indirect Drawer Test" (Figure 13) is performed by placing the palm of one hand over the patella, while extending the index finger of that hand and placing it on the most proximal aspect of the tibial tuberosity. The palm of the other hand is placed on the plantar (caudal) aspect of the metatarsus (bones below the hock). The stifle joint is extended or just slightly flexed, while the hock is flexed to approximately 90 degrees. With a CCLR, the index finger of the hand that is placed around the stifle will move forward. This test frequently allows detection of CCLR in heavily muscled dogs that may not show instability with the "Direct Drawer Test". Sometimes a mild degree of tibial rotation is observed when performing the tests for "Drawer". This should not be mistaken for evidence of CCLR. Dogs with CCLR, especially chronic cases, are at risk for the development of meniscal injuries. The meniscus is a structure that is located between the femur and the tibia. It acts as a cushion between these two bones. Meniscal injuries may occasionally be detected by an audible "click" when the dog is walking or be felt as a "pop" during examination.

Figure 12: "Direct drawer test." The tibia moves cranially with respect to the femur.

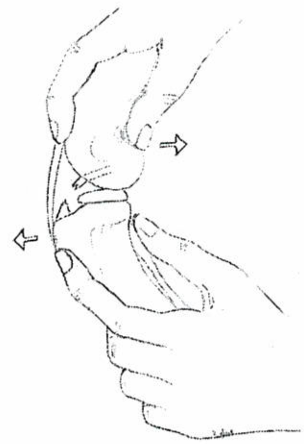
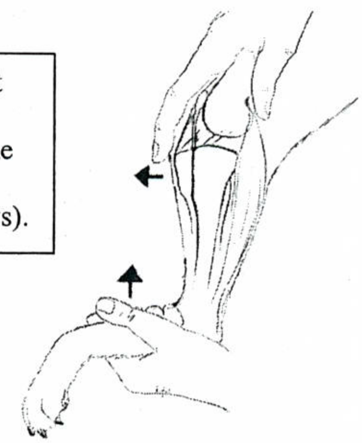


Figure 13: "Indirekt drawer test." While flexing the tarsus the tibia is pushed cranially (red arrows).



Joint swelling without instability of the stifle joint in young dogs may be associated with other etiologies that may need to be diagnosed using more sophisticated tests such as radiography or arthroscopy.
Instability of the collateral ligaments is evaluated by extending the joint and trying to open it laterally or medially. For this test one hand holds the femur above the stifle joint and the other hand holds the tibia below the stifle joint. Then the examiner tries to open the joint medially or laterally. Compare the two legs and be sure to have the hind leg extended. Differences in the degree of joint laxity associated with discomfort are likely the result of collateral ligament damage.
Patella luxations are uncommon in active racing sled dogs. However, they can be the cause of lameness in young dogs that just started training. The tests for patellar luxation should be performed with the stifle in different degrees of flexion and extension. Medial patellar luxation is diagnosed by extending the stifle and partially flexing the tarsus while rotating the later internally (Figure 14). The thumb of one hand is placed on the lateral aspect of the patella, and the patella is pushed medially out of the femoro-patellar joint. The opposite is done for lateral patellar luxations (Figure 15). Dogs with patellar luxations can obtain a secondary CCLR (see above). Therefore it is imperative to always examine dogs with patellar luxations for evidence of CCLR.

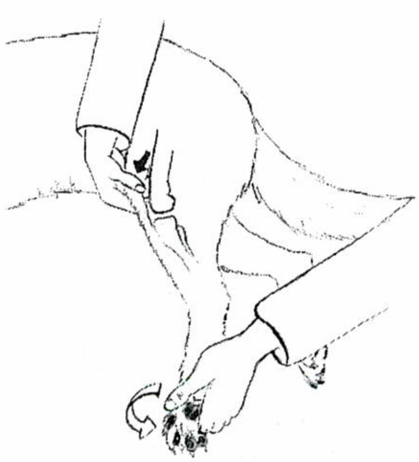
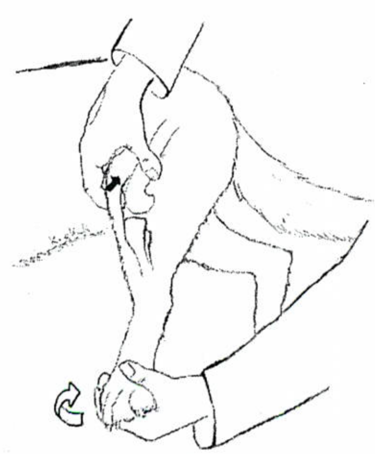


Figure 14 (left): Test for medial patella luxation.

Figure 15 (right) : Test for lateral patella luxation.



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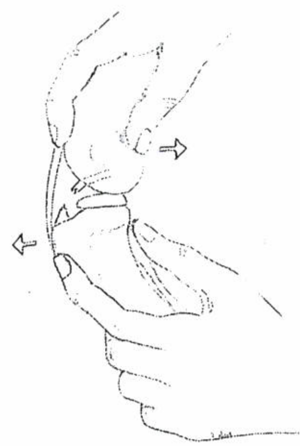

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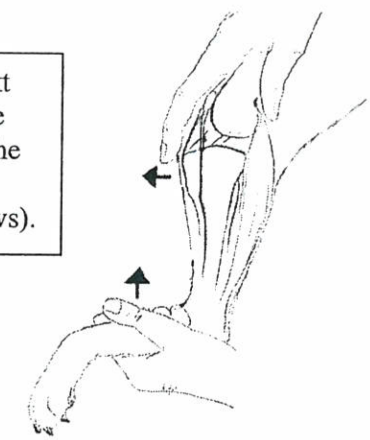

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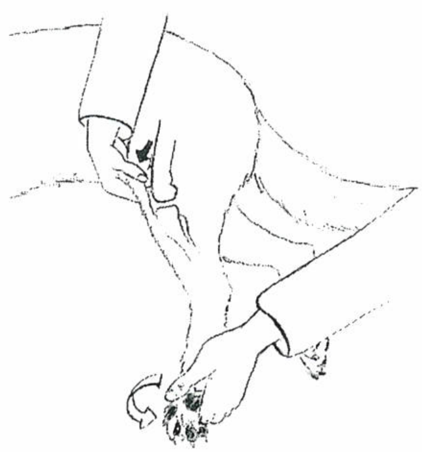
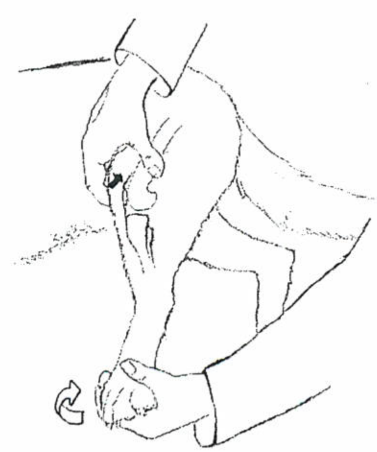


Figure 14 (left): Test for medial patella luxation.

Figure 15 (right) : Test for lateral patella luxation.



9.) Pelvic (hip) injuries of sled dogs are mostly limited to muscle strains. Dogs with hip dysplasia or degenerative changes are unlikely to enter a race on a competitive level. Other injuries such as hip luxations or fractures may occur after accidents on the trail and could be the cause of lameness. Evaluation of the hip joint should begin with simultaneous palpation of both hind legs. Muscle swelling or atrophy can be evaluated by letting both hands glide simultaneously from proximal to distal with one hand on each hind leg. An evaluation for hip luxation or other causes for hip bone asymmetry is done next. The index fingers are placed on on the tuber coxae, the thumbs on the tuber ischii, and the middle fingers on the greater trochanters. Asymmetry of the bones of the pelvis may be an indication of dislocation, fracture, or chronic coxofemoral arthritis. The three fingers of each hand should form a triangle. With the most common form of hip luxation (craniodorsal), the middle finger on the affected side will be positioned more proximally, when compared to the contralateral side (the triangle is "flattened") (Figure 16). Animals with hip luxation frequently are toe-touching lame, and the affected leg is carried in a non-weight-bearing position, with internal rotation of the foot.

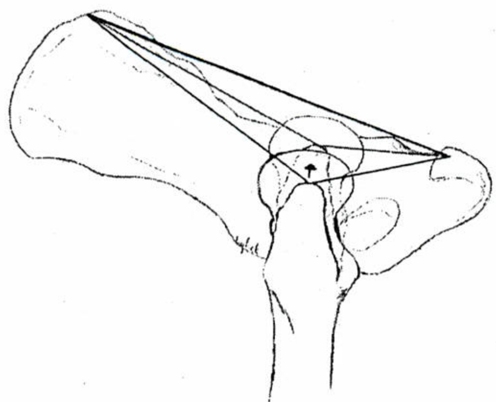


Figure 16: With hip luxation the triangle formed by the examiner's fingers is flattened (red) when compared to normal (black).

Extension of the hip is performed by placing the hand on the femur, just proximal to the cranial aspect of the stifle. The other hand is placed flat on the caudal lumbar spine or hip. The femur is then carefully pulled caudally (Figure 17). A healthy dog should be able to extend the hip joint to a point at which the hind leg is approximately on one level with the hip. Animals with hip injuries may resist this amount of extension. Flexion of the hip is performed by placement of the palm of the examiner's hand just caudal and proximal to the stifle joint, and pulling the femur dorsally and cranially (Figure 17). A consistent method to elicit pain from the hip in animals with a hip joint problem is to abduct the femur. One hand is placed on the medial aspect of the stifle, and the other on the lower back of the animal. Then the leg is carefully pulled laterally. The iliopsoas muscle courses from the transverse processes of 5^{th} to 7^{th} lumbar vertebral bodies to the third trochanter on the femur. To test for iliopsoas muscle injury, the examiner places his arm between the hind legs of the dog, and puts the palm of the hand on the medial aspect of the stifle. The cranial aspect of the hock will automatically rest on the dorsal aspect of the examiner's elbow, while the hind leg is extended. With the fingers of the contralateral hand, the examiner places pressure on the origin of the iliopsoas muscle (Figure 18). A healthy animal will allow this test to be performed without reacting uncomfortably. Fractures of the pelvic bones may be detected during a rectal examination.

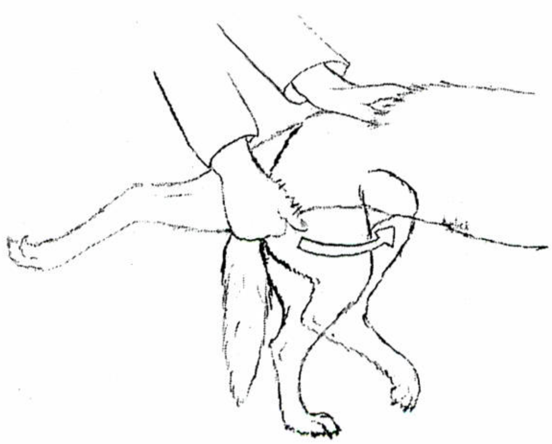


Figure 17: Hip flexion and extension.

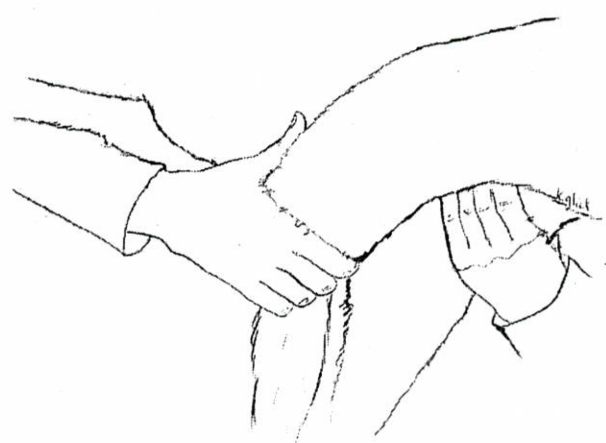


Figure 18: Test for iliopsoas muscle